

BAADE-WESSELINK AND RELATED TECHNIQUES FOR MODE DISCRIMINATION IN NONRADIAL STELLAR PULSATIONS

P. A. STAMFORD and R. D. WATSON

Department of Physics, University of Tasmania, Hobart, Tasmania, Australia

(Received 13 October, 1980)

Abstract. Linearized nonradial expressions for light, colour and radial velocity changes are employed to examine the Baade-Wesselink method as applied to the problem of nonradial mode identification in variable stars. It is demonstrated that the usual assumption of a single-valued relation between surface brightness changes and colour variations fails for spherical harmonics with $l > 1$. A Baade-Wesselink expression taking this into account is then given. Using test models for different variable star regimes it is shown that discrimination between $l = 0, 1$, and 2 is currently feasible provided adequate attention is paid to pressure effects and to the procedure used to obtain radial velocity data. For higher l , mode discrimination, while possible in principle, is constrained by the stringent requirements on radial velocity measurements and by cases of confusion with lower l . A review of related but less observationally demanding possible mode discriminators is given. This demonstrates that techniques based on amplitude and phase information in light and colour data may provide reliable discrimination between $l = 0, 1$, and 2 when pressure effects are allowed for. For these techniques discrimination of higher l is limited by inadequate model atmosphere limb-darkening information and by confusion with lower l .

1. Introduction

Evidence for nonradial stellar pulsations is accumulating in a variety of variable star types. Some δ Scuti stars (Breger, 1979; Balona and Stobie, 1980) and some β Cephei stars (Jerzykiewicz, 1978; Stamford and Watson, 1978; Smith, 1980a) as well as the 53 Persei-type B star variables (Smith, 1978, 1980b; Buta and Smith, 1979), the ZZ Ceti-type white dwarf variables (McGraw, 1978, 1979; Robinson, 1978) and variable A supergiants (Lucy, 1976) have been discussed in this context. However, the basic observational problem of the identification of a nonradial pulsation mode remains a difficult one.

Dziembowski (1977) and Buta and Smith (1979) have derived analytic expressions for the light variations and radial velocity variations resulting from nonradial pulsation in the linear approximation. Dziembowski also suggested that a Wesselink technique could be formulated from these expressions to determine the spherical harmonic order l if the stellar radius R was known, or to get R if l was known. This suggestion was recast in an observationally feasible way by Balona and Stobie (1979a). Application of this method has been subsequently discussed by Balona and Stobie (1979c, 1980).

In this note we consider the two fundamental assumptions of the classical Baade-Wesselink technique in the nonradial pulsation context. These are the assumption of a unique surface brightness-colour index relation (Section 3), and the assumption that the radial velocity variation of the line-forming region can be converted to give the variation in the projected area of the photosphere

(Section 4). As a result of these considerations, it is apparent (Section 5) that the Balona and Stobie (1979a) Baade-Wesselink formula must be modified for $l > 1$ and, in particular, that this modified expression will work for odd l as well as even l . Application of a nonradial Baade-Wesselink method and less observationally demanding variations of it are considered in Sections 5 and 6 by use of test models for different variable star regimes. Difficulties due to pressure effects and limb-darkening uncertainties are also discussed in these sections.

2. Linearized Nonradial Variations

Considering the nonradial pulsation of a spherically symmetric star in the linear approximation, we have, following Dziembowski (1977), a perturbation of the stellar radius

$$\Delta R = \varepsilon R P_{l,m}(\cos \theta) \cos(\omega t + m\phi), \quad (1)$$

where a spherical coordinate system (r, θ, ϕ) is used and the polar axis ($\theta = 0$) corresponds to the stellar rotation axis. R is the equilibrium stellar radius and ε is an amplitude parameter. $P_{l,m}$ are the associated Legendre polynomials.

Similarly it is assumed that the local effective temperature changes, defined in terms of the integrated local emergent flux, can be written as

$$\Delta T_e = B \varepsilon T_e P_{l,m}(\cos \theta) \cos(\omega t + m\phi + \psi_T). \quad (2)$$

That is, it is assumed that the local effective temperature scales with local radius variation, with a phase shift ψ_T between the two. T_e is the equilibrium stellar temperature and B is the ratio of fractional temperature amplitude to fractional radius amplitude. Both B and ψ_T are assumed constant over the stellar surface for a particular pulsation mode. In an adiabatic situation $\psi_T = \pi$ and B would be given explicitly by $B = (1 - \Gamma_2^{-1})C$. As noted by Buta and Smith (1979), C can be taken as

$$C = (4 + \alpha_H^{-1}) - l(l+1)\alpha_H, \quad (3)$$

where

$$\alpha_H = G\rho_\odot Q^2/3\pi = GM/\omega^2 R^3.$$

The stellar mass is M and Q is the pulsation constant.

If the local monochromatic flux is taken to be dependent on a local effective temperature and gravity, we may write for the local fractional monochromatic flux change

$$\begin{aligned} \Delta F_\lambda / F_\lambda &= \left(\frac{\partial \log F_\lambda}{\partial \log T_e} \right) \Delta T_e / T_e + \left(\frac{\partial \log F_\lambda}{\partial \log g} \right) \Delta g / g = \\ &= \alpha_T(\lambda) \Delta T_e / T_e + \alpha_g(\lambda) \Delta g / g, \end{aligned} \quad (4)$$

where $\alpha_T(\lambda)$ and $\alpha_g(\lambda)$ are derived from published model atmospheres, such as

those of Kurucz (1979), at the $(T_e, \log g)$ corresponding to the stellar equilibrium conditions. The application of static atmosphere data to variable conditions must, of course, be viewed with caution, particularly in connection with an effective gravity variation. Here we assumed an effective gravity varying in phase with local gas pressure, $\Delta g/g = p^* \Delta p_g/p_g$, where the coefficient p^* can be obtained from equilibrium stellar atmospheres. Thus,

$$\Delta g/g = -p^* C \Delta R/R. \quad (5)$$

The significant conclusions of this note are not dependent on the details of the gravity term in (4) and results with and without inclusion of this crude model are given. What is apparent, however, is that pressure effects *are* important in general and some model to account for them is required.

Combining Equations (1), (2), and (4) gives

$$\begin{aligned} \Delta F_\lambda / F_\lambda &= \varepsilon P_{l,m} (\cos \theta) \times \\ &\times [\alpha_T(\lambda) B \cos(\omega t + m\phi + \psi_T) - p^* C \alpha_g(\lambda) \cos(\omega t + m\phi)] = \\ &= \varepsilon P_{l,m} (\cos \theta) f_\lambda \cos(\omega t + m\phi + \psi_\lambda), \end{aligned} \quad (6)$$

where

$$f_\lambda = \{[\alpha_T(\lambda) B]^2 + [p^* C \alpha_g(\lambda)]^2 - 2[\alpha_T(\lambda) B p^* C \alpha_g(\lambda)] \cos \psi_T\}^{1/2} \quad (7)$$

and

$$\tan \psi_\lambda = \frac{\alpha_T(\lambda) B \sin \psi_T}{\alpha_T(\lambda) B \cos \psi_T - p^* C \alpha_g(\lambda)}. \quad (8)$$

The f and ψ notation is used here to be consistent with Balona and Stobie (1979a, c). If no effective gravity term is included in (4), then we have simply $f_\lambda = \alpha_T(\lambda) B$ and $\psi_\lambda = \psi_T$.

Integration of the local monochromatic flux change (6) over the stellar disk and retaining only first order terms (Dziembowski, 1977) gives an observed monochromatic flux change

$$\begin{aligned} \Delta m(\lambda, t) &= -1.086 \varepsilon P_{l,m}(\mu_0) [f_\lambda b_{l\lambda} \cos(\omega t + m\phi_0 + \psi_\lambda) + \\ &+ g_{l\lambda} \cos(\omega t + m\phi_0)]. \end{aligned} \quad (9)$$

The angles (i, ϕ_0) denote the direction of the observer and $\mu_0 = \cos i$. The coefficients $b_{l\lambda}$ and $g_{l\lambda}$ are given by Balona and Stobie in terms of integrals involving Legendre polynomials. However, it follows from the properties of Legendre polynomials that these coefficients are simply related by the expression

$$g_{l\lambda} = (2 + l)(1 - l) b_{l\lambda}. \quad (10)$$

Thus only one integral involving the atmospheric limb-darkening is required for the evaluation of $\Delta m(\lambda, t)$. Hence, writing (9) in terms of the wavelength

independent phase shift ψ_T , we have

$$\Delta m(\lambda, t) = -1.086 \varepsilon P_{l,m}(\mu_0) b_{l\lambda} [\alpha_T(\lambda) B \cos(\omega t + m\phi_0 + \psi_T) + \{(2+l)(1-l) - p^* C \alpha_g(\lambda)\} \cos(\omega t + m\phi_0)]. \quad (11)$$

Similarly a colour change, over a wavelength baseline λ_1 to λ_2 , is given by

$$\begin{aligned} \Delta m(\lambda_1, t) - \Delta m(\lambda_2, t) = & -1.086 \varepsilon P_{l,m}(\mu_0) [(\alpha_T(\lambda_1) b_{l\lambda_1} - \\ & - \alpha_T(\lambda_2) b_{l\lambda_2}) B \cos(\omega t + m\phi_0 + \psi_T) + \\ & + \{(2+l)(1-l)(b_{l\lambda_1} - b_{l\lambda_2}) - p^* C(\alpha_g(\lambda_1) b_{l\lambda_1} - \\ & - \alpha_g(\lambda_2) b_{l\lambda_2})\} \cos(\omega t + m\phi_0)]. \end{aligned} \quad (12)$$

A comparison of (11) and (12) demonstrates that, if the $\alpha_g(\lambda)$ terms are omitted, light and colour changes will be in phase for $l=1$, but in general a phase difference exists.

The integration, over a limb-darkened stellar disk, of the surface velocity components seen by a distant observer gives, to first order, a mean observed surface velocity variation of

$$\Delta V_r = \omega \varepsilon R P_{l,m}(\mu_0) (u_{l\lambda} + \alpha_H v_{l\lambda}) \cos(\omega t + m\phi_0 - \pi/2). \quad (13)$$

This expression is used by Dziembowski and Balona and Stobie to compare with observed radial velocity measurements.

If the *mean wavelength* is determined for line profiles and used as the observed radial velocity indicator, as suggested by Balona and Stobie, then such a comparison is acceptable under the circumstances noted in Section 4. The coefficients $u_{l\lambda}$ and $v_{l\lambda}$ appearing in (13) are defined in Dziembowski (1977) and in Balona and Stobie (1979a). They may be expressed in terms of $b_{l\lambda}$ employing recurrence relations of Legendre polynomials. These expressions are

$$u_{l\lambda} = (2l+1)^{-1} [(l+1)b_{l+1\lambda} + lb_{l-1\lambda}] \quad (14)$$

and

$$v_{l\lambda} = l(l+1)(2l+1)^{-1} [b_{l-1\lambda} - b_{l+1\lambda}]. \quad (15)$$

Using $u_{0\lambda} = b_{1\lambda}$ and $v_{1\lambda} = 0$ we therefore find that ΔV_r may be expressed in terms of the same limb-darkening integral $b_{l\lambda}$ as was $\Delta m(\lambda, t)$.

Some example values of the limb-darkening integral $b_{l\lambda}$ are given in Table I. These values are derived from the monochromatic limb-darkening laws tabulated by Kurucz (1969) for equilibrium stellar temperatures and gravities characteristic of the β Cephei, δ Scuti and short-period Cepheid regimes. Although the tabulated values come from the accurate Kurucz limb-darkening solutions, the basic features of Table I can be understood in terms of a linearized limb-darkening law $h(\mu) = \chi_0 + \chi_1 \mu$. In this case the analytic expressions for $b_{l\lambda}$ are those given in Table II, where it has been assumed that $h(\mu)$ is normalized so $\int_0^1 \mu h(\mu) d\mu = 1$. From these expressions it is clear that the $b_{l\lambda}$ values are much

TABLE I
Monochromatic $b_{l\lambda} = \int_0^1 h\mu P_l d\mu$

Regime (T_e , log g) λ	β Cephei (25 000, 4)			δ Scuti (8000, 4)		Short-period Cepheid (6000, 2)		Eddington value b_{lE}
	1500 Å	4000 Å	5500 Å	4000 Å	5500 Å	4000 Å	5500 Å	
$l = 1$	0.719	0.691	0.685	0.723	0.706	0.736	0.713	0.708
2	0.343	0.293	0.281	0.350	0.320	0.374	0.333	0.325
3	0.075	0.033	0.024	0.079	0.055	0.104	0.068	0.062
4	-0.0200	-0.0345	-0.0365	-0.0205	-0.0270	-0.0070	-0.0205	-0.0205
5	-0.0120	-0.0070	-0.0055	-0.0150	-0.0105	-0.0135	-0.0100	-0.0080

less sensitive to the adopted limb-darkening law for $l \leq 2$ than for $l \geq 3$. For the cases shown in Table I, χ_1 is in the range 0.6 to 2.4. As a consequence of this there must be some doubt for $l \geq 3$ in applying monochromatic $b_{l\lambda}$ to normal filter-based observational data when line-blocking is appreciable in the filter passband. The lines will affect the limb-darkening law. Monochromatic values chosen at appropriate effective wavelengths are adequate in relatively line-free passbands. Values of the limb-darkening integral, b_{lE} , which result from the Eddington limb-darkening law ($\chi_1 = 1.5$), are shown in Table I for comparison with the monochromatic values.

In subsequent discussion reference will be made to the amplitude and phase of observational quantities defined in this Section. We use the symbols A_λ , A_{col} and K to refer to the amplitudes of light, colour and radial velocity as defined by Equations (11), (12), and (13) respectively. The symbols ϕ_λ , ϕ_{col} and ϕ_{RV} are used to denote the phases of these quantities relative to $(\omega t + m\phi_0)$, the phase of the sub-observer radius perturbation. Because the bolometric magnitude change, $\Delta m_{\text{bol}}(t)$, which is presented to a distant observer, is not a directly observable quantity we do not refer to it in the following analysis. However, for complete-

TABLE II
Form of $b_{l\lambda}$ for linearized limb-darkening $h(\mu) = \chi_0 + \chi_1\mu$ where $\int_0^1 \mu h(\mu) d\mu = 1$

l	$b_{l\lambda}$
0	1
1	$\frac{2}{3} + \chi_1/36$
2	$\frac{1}{4} + \chi_1/20$
3	$\chi_1/24$
4	$-\frac{1}{24} + \chi_1/72$
5	$-\chi_1/192$

ness we note here the expression for this quantity in the present notation. It is

$$\Delta m_{\text{bol}}(t) = -1.086 \varepsilon P_{l,m}(\mu_0) b_{lE} [4B \cos(\omega t + m\phi_0 + \psi_T) + (2+l)(1-l) \cos(\omega t + m\phi_0)]. \quad (16)$$

The expression is analogous to (9) except that a limb-darkening law applicable to the integrated local emergent flux is required. Hence, b_{lE} is adequate in this context.

3. The Surface Brightness Assumption

If the surface brightness of a star is defined as the observed stellar luminosity per unit projected geometric area A of the photosphere, then the variation of surface brightness $\Delta S(\lambda, t)$ is related to $\Delta m(\lambda, t)$ and the variation in the projected geometric area by

$$\Delta m(\lambda, t) = \Delta S(\lambda, t) - 1.086 \Delta A(t)/A. \quad (17)$$

This is the surface brightness definition adopted by Balona and Stobie (1979a).

Balona and Stobie find, in the linear approximation, that

$$\Delta A(t)/A = \varepsilon P_{l,m}(\mu_0) \cos(\omega t + m\phi_0) 2(2p_l - q_l). \quad (18)$$

It is apparent, however, from the definitions of p_l and q_l and the properties of Legendre polynomials that we have simply

$$(2p_l - q_l) = P_l(0). \quad (19)$$

Thus, as was noted by Balona and Stobie, the projected geometric area variation is zero for odd l in the linear regime.

Combining Equations (11), (17), (18), and (19) we find that

$$\Delta S(\lambda, t) = -1.086 \varepsilon P_{l,m}(\mu_0) [\alpha_T(\lambda) B b_{l\lambda} \cos(\omega t + m\phi_0 + \psi_T) + \{(2+l)(1-l)b_{l\lambda} - p^* C \alpha_g(\lambda) b_{l\lambda} - 2P_l(0)\} \cos(\omega t + m\phi_0)]. \quad (20)$$

From a comparison of (12) with (20) it may be shown that the assumption of a single-valued relation between the surface brightness variations $\Delta S(\lambda_2, t)$ and colour variations $[\Delta m(\lambda_1, t) - \Delta m(\lambda_2, t)]$ is not in general true. This is not surprising when surface gravity changes are included but omitting the $\alpha_g(\lambda)$ terms demonstrates that, even when the monochromatic flux is assumed to be only temperature dependent, a single-valued relation does not exist for $l > 1$. The result follows because equality of the ratio of the amplitudes of the cosine terms in (12) and (20) does not generally hold – i.e.,

$$\frac{\alpha_T(\lambda_2) b_{l\lambda_2}}{[\alpha_T(\lambda_1) b_{l\lambda_1} - \alpha_T(\lambda_2) b_{l\lambda_2}]} \neq \frac{[(2+l)(1-l)b_{l\lambda_2} - 2P_l(0)]}{(2+l)(1-l)(b_{l\lambda_1} - b_{l\lambda_2})}.$$

For $l = 0$ a single-valued relation does occur by virtue of the fact that $b_{0\lambda} =$

$P_0(0) = 1$, making the $\cos(\omega t + m\phi_0)$ amplitude zero and leaving only temperature effects present in the surface brightness and colour changes. For $l = 1$, a similar situation holds, but not for $l > 1$. Estimates of the actual phase difference $(\phi_{S_2} - \phi_{\text{col}})$ between (12) and (20) may be obtained for observed B and $\psi_T = \pi/2$ using the typical variable star regime data discussed in Section 6. These show that, with $\alpha_g(\lambda)$ terms omitted, $(\phi_{S_2} - \phi_{\text{col}})$ remains small for $l = 2$, but is about 15 degrees by $l = 3$ causing substantial departures from a single-valued relation between the surface brightness and colour changes.

In summary, two points should be emphasized. The first is that the existence of a $(\phi_{S_2} - \phi_{\text{col}})$ effect is not dependent on the particular construction of $\Delta F_\lambda/F_\lambda$ in (6). Any wavelength-dependent construction is, in general, going to produce such an effect if $\psi_T \neq \pi$, and, of course, gravity changes will propagate the effect down to the $l = 0, 1$ cases as well. The second point is that, in spite of the problem in relating $\Delta S(\lambda, t)$ and colour changes, a term linearly related to colour *does* appear in the modified Wesselink expression developed in Section 5. Thus, in this sense, the difficulties raised here can be avoided. However, we feel it is important conceptually to note that the idea of a single-valued surface brightness to colour relation must be abandoned in nonradial pulsations.

4. The Radial Velocity Assumption

Difficulties associated with using radial velocity measurements to infer photospheric motion occur because of level effects. These have been extensively considered in work on Baade-Wesselink methods (Parsons, 1972) and are not discussed here. Rather we mention two points which are of particular relevance to nonradial pulsations.

The first point is simply that the velocity variation (13) is wavelength-dependent through the wavelength dependence of limb darkening. Using (14), (15) and Table I data, it can be readily shown that such changes in ΔV_r over the visual are small for low l . However, by $l = 3$ the effect may be of the order of 10% over the visible region and for $l = 4$ changes of up to 25% are possible. This is also a warning that by $l = 3$ it may be expected that the actual dependence of line profiles on their local emergent angle from the atmosphere will be beginning to affect the form of the integrated profile. If a continuum limb-darkening coefficient is used in the $b_{l\lambda}$ appearing in (13), the assumption that ΔV_r may be derived from mean wavelengths of line profiles is clearly insecure for $l > 3$.

The second point is concerned with the question of observationally measuring a quantity which may be related to ΔV_r . For weak lines, small amplitudes and low l , a measurement of the mean wavelength of a line profile may be reasonably assumed to give ΔV_r . Because obtaining the mean wavelength of a profile explicitly requires integration over a digitized spectral output, it is natural to attempt to use a more readily obtainable line characteristic which will approximate the mean wavelength satisfactorily. For $l = 0$ this is simply done, as

can be seen from Figure 1. For higher l the situation is more complex, as is illustrated in Figure 2. In both these figures some typical line characteristics have been used to derive p , the ratio of surface velocity amplitude, v_s , to the observed velocity amplitude, K . In the general nonradial case we define

$$p = v_s/K = \omega \epsilon R P_{l,m}(\mu_0)/K. \quad (21)$$

The line characteristics are plotted as a function of $v_s/v_e \sin i$ to indicate the effect of the rotational velocity field. Values of p using the line centre at different depths (0.33, 0.50, 0.85, 1.0 of line depth) and using the median location are shown. No instrumental broadening has been included. The particular set of model line profiles used to generate Figure 1 and Figure 2 were calculated for β Cephei star conditions by employing an appropriate visual limb-darkening law, taking $v_e = 50 \text{ km s}^{-1}$, and assuming a Gaussian profile with a total broadening of $v_b = 8 \text{ km s}^{-1}$ while covering a range of v_s values. They quantitatively apply only to variables with this v_b/v_e , but qualitatively they illustrate the effects which will be encountered for any pulsating variable due to the interplay of the velocity fields present.

For $l = 0$ it is apparent from Figure 1 that, in the pulsation-dominated regime, p values of line characteristics such as the centre at 0.33 and 0.50 of line depth and

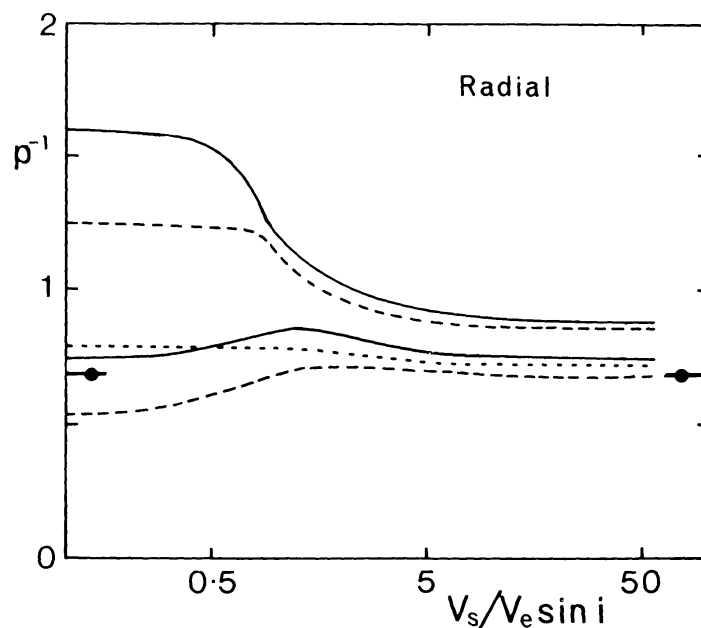


Fig. 1. The ratio of observed velocity amplitude to surface pulsation velocity amplitude, $p^{-1} = K/v_s$, versus $v_s/v_e \sin i$ for a selection of line characteristics from a set of model profiles of a radial pulsation with $v_b/v_e = 0.16$. Profile centre relations at 0.33 depth (dashed line), at 0.50 depth (full line), at 0.85 depth (dashed line) and at 1.00 depth (full line) are shown, with ordinate values increasing along this sequence. The relation using the median wavelength of the profile is a dotted line. The ordinate obtained from using the mean wavelength of the profile is indicated by filled circles.

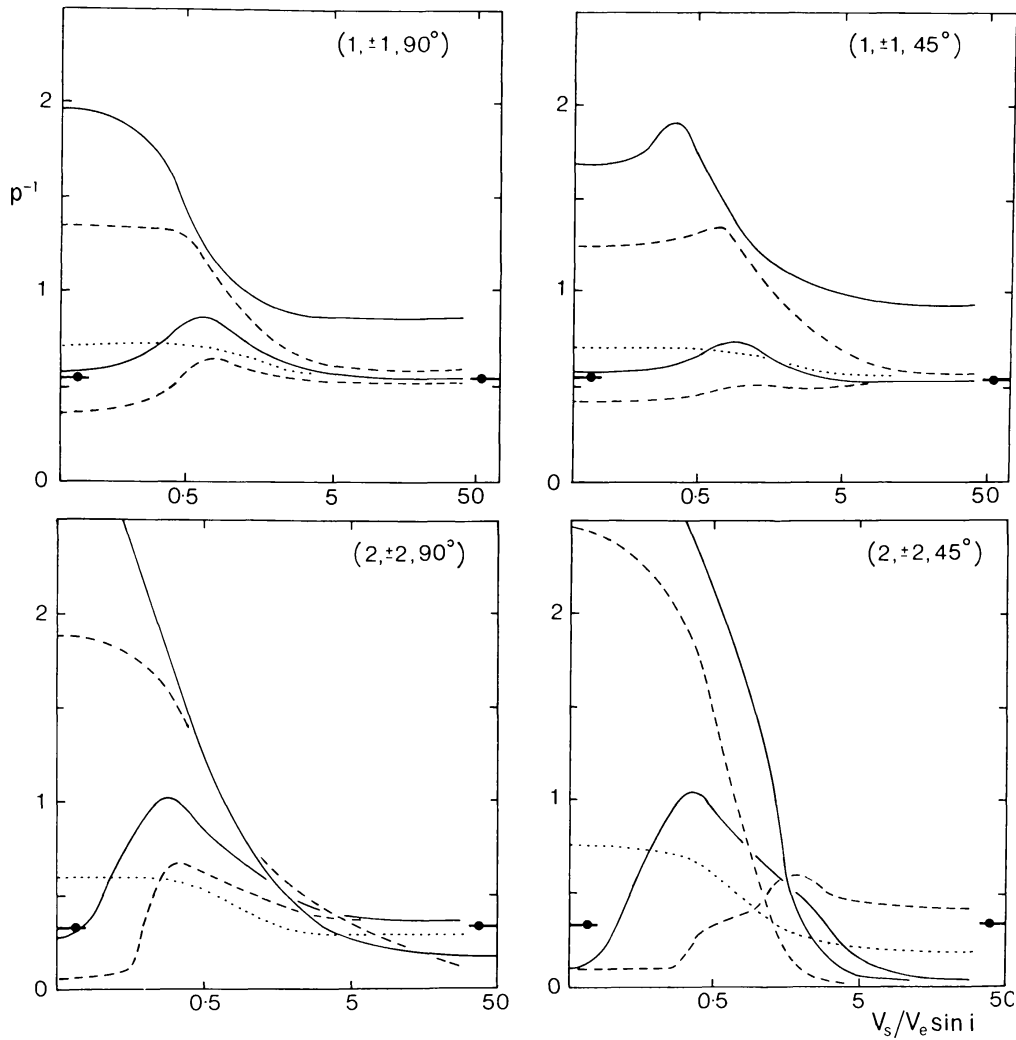


Fig. 2. Symbols are as in Figure 1, but here profiles are from some example nonradial configurations $(l, m) = (1, \pm 1)$ and $(2, \pm 2)$ with $i = 45^\circ$ and 90° .

the median location will not differ greatly from the p value which would have resulted if the mean wavelength of the profile had indeed been measured. In the notation adopted here,

$$p_{\text{mean}} = (u_{l\lambda} + \alpha_H v_{l\lambda})^{-1}. \quad (22)$$

The historically used p value of 1.41 results from (22) using the Eddington limb-darkening law to give $b_{1E} = 0.708$. If we use the $b_{l\lambda}$ values in Table I slightly different p_{mean} result. In Figure 1 the line centre at 0.50 depth gives $p = 1.35$, the median wavelength gives $p = 1.39$ and the line centre at 0.33 depth equals the corresponding p_{mean} of 1.46 because of the quasi-triangular nature of the pulsation-dominated profiles. These values compare favourably with Parsons' (1972) value of 1.31 based on assumed line centre measurements in the line depth range of 0.50 to 0.70. Clearly then, in the radial case, some simply measured line

characteristics can be used to approximate the mean profile wavelength without gross error. However, as a cautionary note, it is also clear that in the regime where the magnitude of the rotational and pulsational velocity fields become comparable it is not satisfactory to use particular p values, such as that of Parsons, if high accuracy is required. This latter point is of relevance to the β Cephei, δ Scuti and short-period Cepheid variables. The effects of rotation on line profiles of radially pulsating stars have been recently discussed by Duval and Karp (1978).

Figure 2 shows that for $l = 1$ the situation is only slightly worse than the radial case, and the mean profile wavelength may be roughly approximated by other simpler line characteristics. For $l = 2$ the situation has become much worse. None of the line characteristics plotted successfully approximates p_{mean} in any regime of $v_s/v_e \sin i$. Hence, in a general nonradial mode identification procedure, it seems that commonly used line characteristics cannot be relied upon to approximate ΔV_r in all circumstances. It is necessary to explicitly find the mean profile wavelength.

5. The Nonradial Baade-Wesselink Expression

Using Equations (11), (12), and the integral of (13) it is possible to derive a Baade-Wesselink expression without having to make any *ad hoc* assumptions concerning the form of the relation between surface brightness and observed colour. The resulting expression is

$$\Delta m(\lambda_2, t) = a_l(\lambda_1, \lambda_2)[\Delta m(\lambda_1, t) - \Delta m(\lambda_2, t)] + 1.086 \beta_l(\lambda_1, \lambda_2) \int \Delta V_r dt/R, \quad (23)$$

where

$$a_l(\lambda_1, \lambda_2) = (\alpha_T(\lambda_1)b_{l\lambda_1}/\alpha_T(\lambda_2)b_{l\lambda_2} - 1)^{-1} \quad (24)$$

and

$$\beta_l(\lambda_1, \lambda_2) = a_l(\lambda_1, \lambda_2)[(2+l)(1-l)(\alpha_T(\lambda_1)/\alpha_T(\lambda_2) - 1) - p^*C(\alpha_g(\lambda_2)\alpha_T(\lambda_1)/\alpha_T(\lambda_2) - \alpha_g(\lambda_1))]b_{l\lambda_1}p_{\text{mean}}. \quad (25)$$

These equations are consistent with the surface brightness to colour relation discussed in Section 3. The points made in Section 4 must be borne in mind in measuring ΔV_r . If a systematic wavelength-dependent effect appears in ΔV_r , it will be assumed for the present discussion that ΔV_r , and hence p_{mean} , are evaluated at λ_2 .

At its most basic level, the difference between the present work and that of Balona and Stobie (1979a) can be gauged by examining (25) with effective gravity terms omitted and the wavelength dependence of limb-darkening ignored. Such a comparison is made in Table III using the Eddington limb-darkening integral in

TABLE III
Comparison of Baade-Wesselink values of β_l/p_{mean}

l	β_l/p_{mean}	
	$2P_l(0)$ Balona and Stobie (1979a)	$(2+l)(1-l)b_{lE}$ Equation (25) with Eddington limb-darkening and no effective gravity terms
0	2	2
1	0	0
2	-1	-1.30
3	0	-0.62
4	0.75	0.37
5	0	0.22

(25). It is apparent from Table III that mode discrimination is possible, in principle at least, for all l and not just even values.

5.1. EXAMPLE OF BAADE-WESSELINK COEFFICIENTS

To evaluate $a_l(\lambda_1, \lambda_2)$ and $\beta_l(\lambda_1, \lambda_2)$ and pursue mode discrimination possibilities, information on $\alpha_T(\lambda)$, $\alpha_g(\lambda)$, $b_{l\lambda}$ and p^*C is required. We have chosen three test models to do this. A model with $(T_e, \log g) = (25\,000, 4)$ serves to illustrate the case of β Cephei and other nearby B star variables. A $(8\,000, 4)$ model applies to the δ Scuti regime and a $(6\,000, 2)$ model represents the short-period Cepheid situation. The latter case was included, not because there is any current evidence for nonradial pulsations in this regime, but to provide, together with the δ Scuti model, a pair of examples bracketing variables in the lower reaches of this instability region. B and V passbands were selected to illustrate the δ Scuti and short-period Cepheid cases. For the β Cephei example, a $1500\text{ \AA}$ passband was also selected. This wavelength region is covered by satellite ultraviolet data. Both $(1500, V)$ and (B, V) cases were carried through in the β Cephei example to demonstrate their extreme differences.

Data used in the test models are given in Table IV. The $\alpha_T(\lambda)$ and $\alpha_g(\lambda)$ values for the various passbands come from Kurucz (1979). The p^*C numbers are meant as approximate values only. They were derived using p^* values taken from the photospheric regions of the appropriate Kurucz models. These are $p^* \sim 1.3 - 1.5$. The C values come from typical Q . These were taken to be 0.027 in the β Cephei model (Balona and Feast, 1975; Shobbrook, 1978), 0.022 in the δ Scuti model (Breger and Bregman, 1975) and 0.040 in the short-period Cepheid model (Cox, 1979). This δ Scuti value corresponds to overtone pulsation, as suggested for $T_e > 7800\text{ K}$ by Breger and Bregman. For cooler δ Scuti variables a $Q \sim 0.030$ is more appropriate and this reduces p^*C to ~ 25 . The presence of

TABLE IV
Data for test models in different variable star regimes

Regime (T_e , log g)	β Cephei (25 000, 4)			δ Scuti (8000, 4)		Short-period Cepheid (6000, 2)	
Passband	1500 Å	<i>B</i>	<i>V</i>	<i>B</i>	<i>V</i>	<i>B</i>	<i>V</i>
$\alpha_T(\lambda)$	3.85	2.00	1.85	5.05	3.75	6.40	4.90
$\alpha_g(\lambda)$	+0.002	-0.027	-0.024	-0.043	-0.018	-0.013	-0.014
p^*C		30		40 ^a		15	
(B_{observed} , ψ_T)		(2.5, 180°)		(2.5, 90°)		(1.5, 90°)	
$B_{\text{adiabatic}}$		5.5		12.5 ^a		4.5	
$\omega R/1.086 \text{ km s}^{-1}$		2500		2000 ^a		500	

^a These values are for $Q \sim 0.022$ and reduce to $p^*C \sim 25$, $B_{\text{adiabatic}} \sim 7.5$, $\omega R/1.086 \sim 1500 \text{ km s}^{-1}$ for a $Q \sim 0.030$ which approximates a radial fundamental pulsation.

different modes in the regimes of the other test models would affect their p^*C similarly. Equation (3) implies that the C values are l dependent. The effect is only appreciable in the short-period Cepheid example. The Table IV value is an intermediate one between the $l = 0$ and 5 results in this case.

A limited check on the adopted effective gravity model is possible for the δ Scuti test example, because gravity and radius changes around the pulsation cycle are given directly for AD CMi by Breger (1975) and for RS Gru by McNamara and Feltz (1976). This check shows that the phase of the effective gravity changes in the model is roughly correct, as is the amplitude, expressed by (5), provided the assumption is made that AD CMi and RS Gru have Q appropriate to fundamental pulsations. The Table IV p^*C value may be a little high, but not by more than a factor of two. No direct check on the effective gravity model is possible in the β Cephei case, though indirect evidence (Stamford and Watson, 1978) supports the existence of an effect of the order of that given by (5). Observed effective gravity changes in Cepheids cannot be validly compared with (5) because of their highly nonlinear situation. Clearly only very small radius excursions are permitted by the Table IV p^*C coefficients before nonlinear effects occur. The best that can be said for the short-period Cepheid example is that perhaps (5) will give a guide to the importance of pressure effects in the low amplitude limit.

The Baade-Wesselink coefficients derived for the test models are given in Table V. The corresponding p_{mean} are also given. The $b_{l\lambda}$ used in deriving Table V data come from Table I by interpolating at appropriate effective wavelengths. Values of $\beta_l(\lambda_1, \lambda_2)$ both with and without effective gravity changes are included in Table V. These show that pressure effects may be substantial in the β Cephei and δ Scuti regimes for low l . The importance of effective gravity changes on

TABLE V
Example Baade-Wesselink coefficients

Regime	β Cephei (1500, V)			β Cephei (B, V)			δ Scuti (B, V)			Short-period Cepheid (B, V)		
	a_l	β_l		a_l	β_l		p_{mean}	a_l	β_l	p_{mean}	a_l	β_l
p^*C	0	30		0	30		0	40		0	15	
$l=0$	0.9	2.9	5.1	12	2.9	2.6	1.46	2.9	2.8	-0.2	1.42	3.2
1	0.8	0	1.8	11	0	-0.3	1.84	2.8	0	-2.5	1.76	3.0
2	0.6	-2.9	-1.9	8	-2.5	-2.7	3.05	2.4	-3.1	-4.8	2.82	2.5
3	0.2	-1.2	-1.0	2.5	-0.5	-0.5	7.9	1.4	-2.3	-2.7	6.6	1.4
4	7	260	240	26	130	130	90	6.4	35	39	39	-6.2
5	0.3	-5.3	-5.1	2.5	-2.3	-2.3	-55	1.4	-14	-15	-78	1.5

Note: The p_{mean} are here evaluated at V .

Wesselink analyses of δ Scuti stars has been recognised by Breger (1975) who uses gravity-corrected colours in studying radial pulsations.

5.2. MODE-DISCRIMINATION ABILITY

The principal mode discrimination ability of the nonradial Baade-Wesselink coefficients comes from the $(2+l)(1-l)$ term in (25), though by $l=3$ the l dependence of $b_{l\lambda}$ is affecting both coefficients strongly. To assess the practical l discrimination possibilities we require, in addition to knowing that pressure effects must be included, some idea of the uncertainties related to the $\alpha_T(\lambda)$ and $b_{l\lambda}$ values used in (24) and (25). These are difficult to obtain. However, the $a_0(B, V)$ for the short-period Cepheid and δ Scuti cases does offer an important point of contact with observation which gives some information on $\alpha_T(\lambda)$. The $a_0(B, V)$ may be identified with the surface brightness to colour gradient term appearing in maximum likelihood Wesselink analyses. It is well known that observed values of this quantity vary little along the Cepheid instability strip and Balona (1977) determines a mean value of 2.15 ± 0.02 . Strictly speaking, we should here make a comparison with values determined from the fundamental sinusoidal components of the Cepheid variations, as given by Balona and Stobie (1979c). For short-period Cepheids, Balona and Stobie (1979d) note that $a_0(B, V) = 2.5 \pm 0.04$. The difference between this and the $a_0(B, V) = 3.2$ listed in the Table V short-period Cepheid test model implies an observed $\alpha_T(B)/\alpha_T(V)$ some 10% larger than given by the Kurucz (1979) models. The Bell and Gustafsson (1978) models, on the other hand, give $a_0(B, V) < 2.5$, which is in closer agreement with the observations. Both sets of models, however, agree in predicting only small changes of $a_0(B, V)$ along the Cepheid instability strip. This can be seen in Figure 3 where model contours of $a_0(B, V)$ are plotted on the $(T_e, \log g)$ plane. From this figure it is also seen that a range of $a_0(B, V)$ is possible in the δ Scuti regime. Observed values of $a_0(B, V)$ for RS Gru (Balona and Martin, 1978) and the radial mode in 1 Mon (Balona and Stobie, 1980) fall within the hatched δ Scuti region in Figure 3, though at somewhat higher T_e than the average T_e observed for these particular stars.

As a result of the preceding considerations it seems prudent to assume that errors of at least ± 0.1 may be present in model $\alpha_T(\lambda)$, quite apart from any uncertainty introduced when assigning equilibrium $(T_e, \log g)$ values to a particular star. Errors of twice this are necessary to account for the discrepancy in the Cepheid region. For $b_{l\lambda}$ we assumed that errors equivalent to a change of χ_1 in Table II by ± 0.1 represented minimum uncertainties. Such an error in $b_{l\lambda}$ may be introduced by the application of monochromatic $b_{l\lambda}$ to any filter-based data which suffers a relatively small amount of line-blocking. Errors several times this size may result from applying monochromatic $b_{l\lambda}$ to β Cephei 1500 Å and short-period Cepheid B passbands, where line-blocking is in the vicinity of 20%.

To assess the limitations in mode discrimination ability caused by minimum uncertainties in the Table V Baade-Wesselink coefficients, we therefore deter-

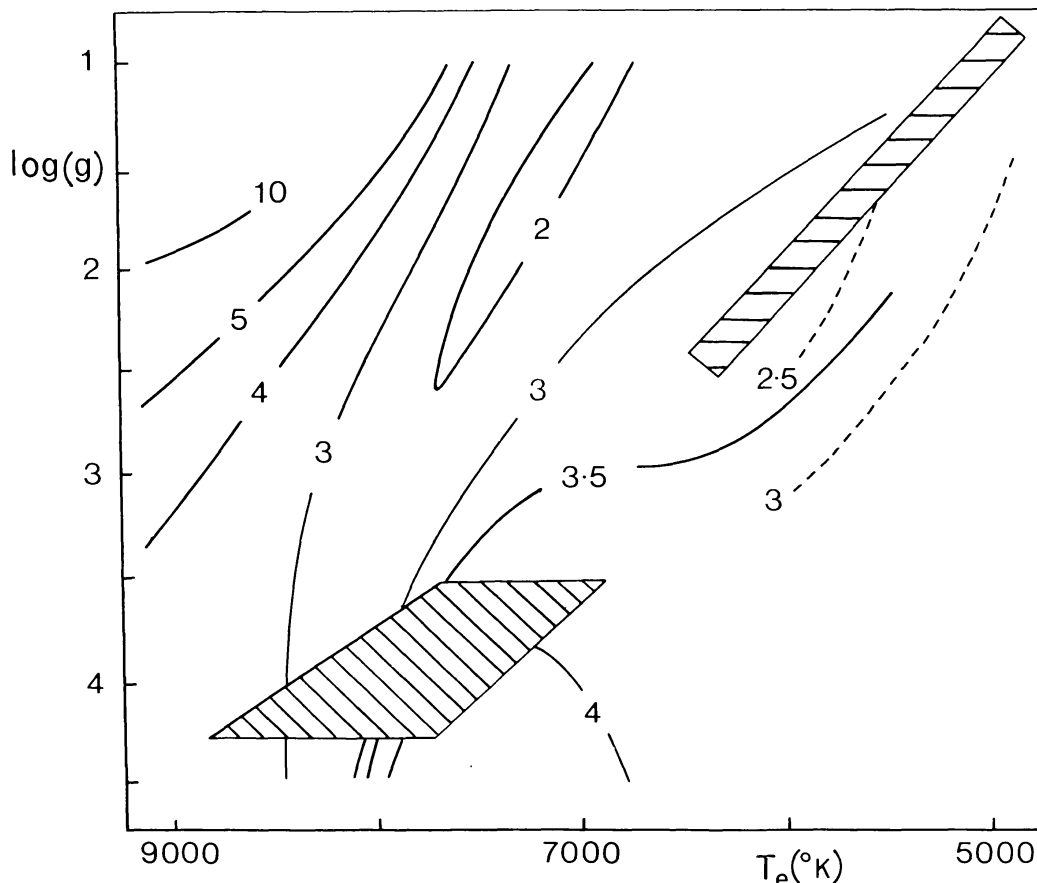


Fig. 3. Contours of the Baade-Wesselink coefficient $a_0(B, V)$ in the $(T_e, \log g)$ plane. Values derived from Kurucz (1979) model-atmosphere data are shown as full lines and those derived from Bell and Gustafsson (1978) are shown as dashed lines. The lower hatched area is the δ Scuti region according to Breger and Bregman's (1975) work. The upper hatched area gives the location of the Cepheid instability strip as summarized by Cox (1979).

mined the range in both of these coefficients which resulted from independently varying $\alpha_T(\lambda_1)$, $\alpha_T(\lambda_2)$, and χ_1 in each of $b_{l\lambda_1}$ and $b_{l\lambda_2}$, by ± 0.1 . It was clear from this that $a_l(\lambda_1, \lambda_2)$ is not a practical mode discriminant in any test regime. The errors in $\alpha_T(\lambda)$ present the problem for $l \leq 2$ while for $l \geq 4$ the $b_{l\lambda}$ errors dominate. It was equally clear that $\beta_l(\lambda_1, \lambda_2)$ is expected to be a successful mode discriminant for $l \leq 2$ provided due attention is paid to pressure effects. This is true in all test regimes except the β Cephei (B, V) case, which is completely unreliable because of its small $[\alpha_T(\lambda_1) - \alpha_T(\lambda_2)]$. For higher l the mode discriminating ability of $\beta_l(\lambda_1, \lambda_2)$ is far less certain. Large positive or negative $\beta_l(\lambda_1, \lambda_2)$ values would distinguish $l = 4$ or 5 from smaller l . However, $l = 3$ is not readily distinguishable from $l = 2$ in any case, and may also be confused with $l = 1$ in the δ Scuti example.

5.3. APPLICATION OF THE BAADE-WESSELINK EXPRESSION

Here we consider appropriate means of applying (23) to observational data. This is done in general terms and then applied to practical $l \leq 2$ examples in Section 5.4.

In the linear regime there are only six observational quantities required to apply a Baade-Wesselink technique for a particular mode. These are the amplitude and phase of the light, colour and radial velocity variations. Writing (23) in terms of these we find that

$$A_{\lambda_2} \cos(\omega t + m\phi_0 + \phi_{\lambda_2}) = a_l(\lambda_1, \lambda_2) A_{\text{col}} \cos(\omega t + m\phi_0 + \phi_{\text{col}}) + [1.086\beta_l(\lambda_1, \lambda_2)/\omega R] K \sin(\omega t + m\phi_0 + \phi_{RV}). \quad (26)$$

Hence, in terms of observational quantities, the Baade-Wesselink coefficients (24) and (25) may be expressed as

$$a_l(\lambda_1, \lambda_2) = \frac{A_{\lambda_2} \cos(\phi_{\lambda_2} - \phi_{RV})}{A_{\text{col}} \cos(\phi_{\text{col}} - \phi_{RV})} \quad (27)$$

and

$$\beta_l(\lambda_1, \lambda_2) = -\frac{\omega R}{1.086} \frac{A_{\lambda_2}}{K} \frac{\sin(\phi_{\lambda_2} - \phi_{\text{col}})}{\cos(\phi_{\text{col}} - \phi_{RV})}, \quad (28)$$

unless $(\phi_{\text{col}} - \phi_{RV})$ has values of $\pm\pi/2$.

In the δ Scuti and short-period Cepheid cases, observations typically give light and colour changes π out of phase with radial velocity variations. In this circumstance one may certainly obtain independent values of $a_l(\lambda_1, \lambda_2)$ and $\beta_l(\lambda_1, \lambda_2)$ from the observations in order to compare with model values given by (24) and (25). In principle one could have hoped to solve for both l and R independently from (27) and (28). However, the practical limitations on the mode discrimination ability of $a_l(\lambda_1, \lambda_2)$, noted in Section 5.2, preclude the current use of (27). Hence we are limited to using (28) in those circumstances where the mode discrimination ability of $\beta_l(\lambda_1, \lambda_2)$ is not destroyed by having only a rough estimate of R . Certainly for $l \leq 2$ one will still be able to mode discriminate.

In the β Cephei case, observations typically give light and colour changes lagging by $\pi/2$ with respect to radial velocity variations. In this situation $a_l(\lambda_1, \lambda_2)$ and $\beta_l(\lambda_1, \lambda_2)$ are not independently obtainable from the observations via (27) and (28) but a linear relation exists between them. Equation (26) gives this relation as

$$\beta_l(\lambda_1, \lambda_2) = \frac{\omega R}{1.086} \frac{A_{\lambda_2}}{K} \left[a_l(\lambda_1, \lambda_2) \frac{A_{\text{col}}}{A_{\lambda_2}} - 1 \right]. \quad (29)$$

Thus model values of $a_l(\lambda_1, \lambda_2)$ must be used together with the observations to solve for $\beta_l(\lambda_1, \lambda_2)$ and hence mode discriminate. Equation (29), under another guise, has already been used for mode discrimination purposes in β Cephei stars by Stamford and Watson (1980), when plotting K/A_V versus A_{1500-V}/A_V data for comparison with model predictions. As will be seen in Section 6, it is the value of $[a_l(\lambda_1, \lambda_2)A_{\text{col}}/A_{\lambda_2} - 1]$ which is of *prime* observational importance in deter-

mining the l dependence of (29) for $l \leq 2$. Here $a_l(\lambda_1, \lambda_2)$ does not change grossly, and the main change is via the l dependence of $A_{\text{col}}/A_{\lambda_2}$. Contrast this with the δ Scuti and short-period Cepheid test cases where it is the value of $(\phi_{\lambda_2} - \phi_{\text{col}})$ which is of *prime* observational importance in determining the $l \leq 2$ dependence of (28). This is also seen in Section 6. Finally, it is worth noting that, when using (28) or (29) for mode discrimination purposes, the large scatter of $\beta_l(\lambda_1, \lambda_2)$ with l means that the method is not as highly demanding of phase matching between light or colour data and radial velocity observations which are not simultaneous, as is a normal radial Wesselink analysis aiming for a precise R determination.

5.4. EXAMPLE APPLICATIONS

In Figure 4 application of the nonradial Baade-Wesselink technique for mode discrimination is illustrated in each of the variable star test regimes. Observations are compared with $\beta_l(\lambda_1, \lambda_2)$ from either (28) or (29) for $l \leq 2$ and pressure effects are shown.

For examples of the β Cephei regime we firstly examine the four cases, γ Peg, δ Cet, β Cep and the large amplitude mode of σ Sco. All of these cases have been independently identified, from a study of line profile variations by Campos and Smith (1980), as being radial pulsators. Observed $\beta_l(1500, V)$ were derived from (29) using data sources listed by Stamford and Watson (1977, 1978). An $a_l(1500, V) = 0.8 \pm 0.2$, to adequately cover all $l \leq 2$ cases, and an $R = 8R_\odot$ were initially assumed. The effect of observational errors in light, colour and velocity amplitudes, together with these $a_l(1500, V)$ errors, is shown in Figure 4 where the mean value for these four stars is plotted. This clearly agrees with the model $\beta_0(1500, V)$ value. Each of the four stars individually favours an $l = 0$ identification, though with appropriately larger error ranges. The effect of unknown R values was then assessed by permitting R to vary from $5R_\odot$ to $11R_\odot$. The increased range in the observed $\beta_l(1500, V)$ due to this is also shown in Figure 4. Clearly identification with $l = 0$ remains secure and a pressure effect of roughly the same order as given by $p^*C = 30$ is indicated. As a second illustration of the β Cephei regime we examined β CMa, for which Beeckmans and Burger (1977) give satellite ultraviolet data for the two principal pulsation modes. The observed $\beta_l(1500, V)$ for these modes are identified in Figure 4 by the Struve notation P_1 (short period) and P_2 (long period). These $\beta_l(1500, V)$ values were obtained in the same manner as described in the first example. They suggest that the P_2 mode is an $l = 2$ pulsation while the P_1 mode may be either $l = 0$ or 1. The radial velocity uncertainties discussed in Section 4 will apply to the P_2 mode if indeed it is an $l = 2$ pulsation. To this must be added the uncertainty, noted by Smith (1980a), that in nonradial multimode pulsations the radial velocity results are not always additive when multiple modes interact, as is assumed in periodogram analyses.

The best illustration of the δ Scuti regime is afforded by the excellent 1 Mon

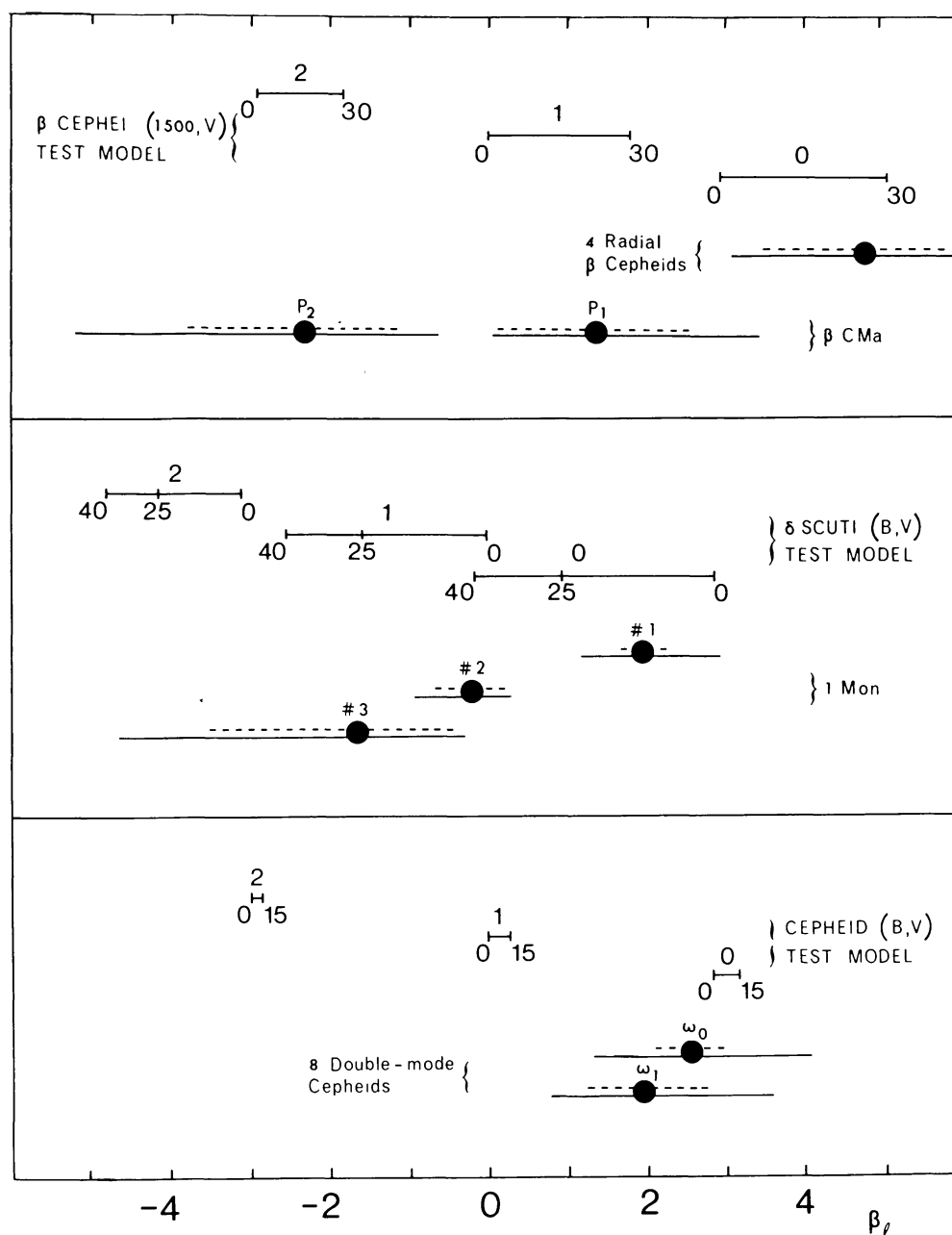


Fig. 4. Observed and predicted Baade-Wesselink $\beta_l(\lambda_1, \lambda_2)$ values for $l = 0, 1, 2$. The corresponding $p \cdot C$ values are shown below fiducial marks on the model lines. Observations to illustrate each of the variable star regimes come from data listed by Stamford and Watson (1977, 1978) and Balona and Stobie (1979b, 1980). The period designations are those given in these sources and are discussed in the text. Dashed error bars show the range of $\beta_l(\lambda_1, \lambda_2)$ permitted by the observational errors if radii are specifically taken as $8 R_\odot$, $3 R_\odot$, and $40 R_\odot$ in the β Cephei, δ Scuti and short-period Cepheid regimes respectively. Full error bars indicate how these ranges are extended if respective radii are permitted to vary from $5 R_\odot$ to $11 R_\odot$, $2 R_\odot$ to $4 R_\odot$, and $25 R_\odot$ to $55 R_\odot$ in these variable star regimes.

data of Balona and Stobie (1980). Values of $\beta_l(B, V)$ were obtained from this data using (28) and an R initially assumed to be $3R_\odot$. Mode #1 was identified by Balona and Stobie as a radial pulsation, while they suggested that the next best observed modes, #2 and #3, were $l = 1$ pulsations. These conclusions were based on an examination of the phase difference between light and colour data, a technique which we discuss in Section 6. Figure 4 shows that the direct Baade-Wesselink technique is consistent with this interpretation. Mode #1 is certainly radial, even if a liberal range of possible R , $2R_\odot$ to $4R_\odot$, is employed. A pressure effect is indicated, though perhaps less than the $p \cdot C = 25$ value which is applicable in our model if the 1 Mon Q value of Breger and Bregman (1975) is used. Mode #2 is an $l = 1$ pulsation according to Figure 4, but this figure alone does not distinguish between $l = 1$ and 2 for mode #3. Other arguments outlined by Balona and Stobie are necessary to decide in favour of the $l = 1$ choice for mode #3.

To illustrate the short-period Cepheid regime we relied on the data for eight double-mode Cepheids given by Balona and Stobie (1979b). They provide information on the fundamental sinusoidal variations of light, colour and radial velocity for both the fundamental period, ω_0 , and the first overtone, ω_1 . Mean $\beta_l(B, V)$ for these eight stars, calculated using (28), are shown in Figure 4. Both ω_0 and ω_1 values identify with radial pulsations even when R is permitted to vary over a range from $25R_\odot$ to $55R_\odot$ about an initially assumed value of $40R_\odot$. Data for individual stars favour the $l = 0$ identification though with considerably larger error ranges. The Figure 4 plot for these eight stars suggests, in agreement with our model, that no large pressure effect exists in this regime.

The general conclusion that may be drawn from Figure 4 is that practical mode discrimination between $l = 0, 1$, and 2 is possible in all test regimes. Observational data plotted in this figure suggests discrimination against $l = 4$ and 5 coefficients listed in Table V with only one exception. However, the $l = 3$ case remains confused, as mentioned in Section 5.2.

6. Related Amplitude and Phase Techniques

The principal advantage of the Baade-Wesselink technique for l discrimination is that, once the observations of the variations have been acquired, the only extra information needed is an approximate value of R and a knowledge of stellar equilibrium conditions to give the required model atmosphere quantities. Its disadvantage is that observations of light, colour and radial velocity are *all* required. Moreover those observations are generally required in multimode situations and situations where the variations are often very small. It is therefore clearly desirable to investigate whether less observationally demanding techniques for l discrimination are possible.

By considering amplitude ratios of, or phase differences between, light colour and radial velocity variations, one can obtain observational quantities which rely

on subsets of the Baade-Wesselink observational requirements while remaining free of the unknown amplitude parameter $\varepsilon P_{l,m}(\mu_0)$. The price paid for this reduced observational requirement is the necessity to know values for the fundamental parameters (B, ψ_T) appearing in (11) and (12). Here we review all the possible amplitude ratio and phase difference combinations in each of the test model regimes after justifying our choice of approximate (B, ψ_T) values for these regimes. The use of $A_{\text{col}}/A_{\lambda_2}$ in the β Cephei regime (Stamford and Watson, 1978; or Balona and Stobie, 1979c) and of $(\phi_{\lambda_2} - \phi_{\text{col}})$ in the δ Scuti regime (Balona and Stobie, 1979c, 1980) appears naturally within the context of this review.

6.1. TEST MODEL (B, ψ_T) VALUES

For radial pulsations it is observationally apparent that $\psi_T \sim \pi$ in the β Cephei regime while it is much closer to $\pi/2$ for variables in the Cepheid instability strip. The work of Balona and Stobie (1979c) in fact suggests $90 < \psi_T < 140^\circ$ for classical Cepheids. Similar values are anticipated for the δ Scuti stars (Balona and Stobie, 1980).

In the radial pulsation case an approximate observed B value follows from the fact that $\eta \sim 0.1$ in a wide range of variable star regimes (Watson, 1971). Here $\eta = \varepsilon/A_{\text{bol}}$, the ratio of fractional radius to bolometric magnitude amplitudes. Using this η , together with $l = 0$ and $\psi_T = \pi$ or $\pi/2$ in (16), it is apparent that a $B \sim 2.5$ applies to our test regimes. Recent more detailed work confirms this approximate value. Stamford and Watson (1978) derive observed B for radially pulsating β Cephei stars which range from 1.6 to 3.2. A similar range of observed B values results from the work of Breger, Hutchins and Kuhl (1976) on δ Scuti radial pulsators. The classical Cepheid work of Balona and Stobie (1979c) implies observed B values which range from 1.2 to 2.0 at a period corresponding to that of our short-period Cepheid test model. The choice of Table IV test model observed B values comes from these considerations.

In Table IV adiabatic B estimates are also listed. These were obtained by employing (3) with $l = 0$, and using the same Q as yielded the p^*C estimates discussed in Section 5.1. Observed B are down by factors of between two and five on their corresponding adiabatic values. A further interesting point is that there is clear evidence in the classical Cepheid strip (Balona and Stobie, 1979c), and some weak evidence in the case of the β Cephei strip (Stamford and Watson, 1978), that the ratio of observed to adiabatic B decreases in going up the respective instability strips. For the classical Cepheids this ratio drops from ~ 0.5 to ~ 0.2 and for the β Cepheids it falls from ~ 0.5 to ~ 0.25 . Thus either the adiabatic assumption or the use of (3) is becoming increasingly invalid at higher luminosities. Alternatively, nonlinear effects may contribute to the phenomenon. Available linear nonadiabatic theoretical pulsation models of δ Scuti stars (Stellingwerf, 1979) and β Cephei stars (Davey, 1970; Saio and Cox, 1980) give values of this B ratio which are larger than those observed. In the β Cephei

models values of this ratio are ~ 1.0 . This suggests that it may indeed be the use of a boundary condition based on (3) which is presenting the problem because of dissipation of mechanical energy in travelling waves.

For nonradial pulsation very little information on (B, ψ_T) is available. Dziembowski (1977) comments that, in his linear nonadiabatic δ Scuti models, B depends mainly on ω for low l and varies little with l itself. His models also suggest that any ψ_T changes are less than 40 degrees for low l . The 1 Mon analysis of Balona and Stobie (1980) certainly indicates that the $l = 0$ and $l = 1$ (B, ψ_T) values in this star do not greatly differ. The Saio and Cox β Cephei models similarly indicate that, for a given ω , B does not change grossly with l , and that ψ_T in general remains close to the radial value of $\sim \pi$.

In view of this, we took the radial values of (B, ψ_T) listed in Table IV as also being indicative of values for low l nonradial pulsation in our test models. This will be adequate provided any mode discriminator considered requires only an approximate knowledge of (B, ψ_T) .

6.2. USEFUL MODE DISCRIMINATORS

Using equations (11), (12), and (13), values of $A_{\text{col}}/A_{\lambda_2}$, K/A_{λ_2} , $(\phi_{\lambda_2} - \phi_{\text{col}})$ and $(\phi_{\lambda_2} - \phi_{RV})$ were generated for each of the test model regimes by employing the same Table IV and $b_{l\lambda}$ data used to generate the example Baade-Wesselink coefficients. These data were supplemented by the observed (B, ψ_T) values in Table IV and, in the case of K/A_{λ_2} , by the $\omega R/1.086$ scale factors. The latter scale factors are illustrative approximations only. In δ Scuti variables for example, $\omega R/1.086$ may vary from the listed value of ~ 2000 for a $\log g = 4$ first overtone pulsator to ~ 1000 for a $\log g = 3.7$ fundamental pulsator. The results of the amplitude ratio and phase difference calculations are presented in Table VI and pressure effects are shown. The l dependence of each of the Table VI data sets was then examined in turn. If mode discrimination seem possible then the effect of poorly known (B, ψ_T) was investigated to see if discrimination remained practical under this limitation.

For $l \leq 2$, the $A_{\text{col}}/A_{\lambda_2}$ values in Table VI suggest that this ratio may be useful in the β Cephei regime but not in the δ Scuti and short-period Cepheid test cases. The principal source of this l dependence in the β Cephei test model is the $(2+l)(1-l)$ term in (11), which determines the 'geometric' contribution to the light amplitude A_V . For $l = 0$ and 2 the magnitude of this geometric term in each of the test models is a substantial fraction of the dominant contributor to A_V , the 'temperature' term $\alpha_T(V)B$. The l dependence introduced by the geometric term in A_V can be seen by noting that, in the absence of pressure effects, $A_{\text{col}}/A_{\lambda_2}$ may be approximated by

$$\frac{A_{\text{col}}}{A_{\lambda_2}} \sim \frac{(\alpha_T(\lambda_1) - \alpha_T(\lambda_2))B}{\{[\alpha_T(\lambda_2)B]^2 + [(2+l)(1-l)]^2 + 2\alpha_T(\lambda_2)B(2+l)(1-l)\cos\psi_T\}^{1/2}} \quad (30)$$

TABLE VI
Example amplitude ratios and phase differences for test models

$A_{\text{col}}/A_{\lambda_2}$		$K/A_{\lambda_2} \text{ km s}^{-1} \text{ mag}^{-1}$									
Regime (B, ψ_T)	p^*C	Short-period									
		β Cephei (1500, V) (2.5, 180°)	β Cephei (B, V) (2.5, 180°)	δ Scuti (B, V) (2.5, 90°)	Cepheid (B, V) (1.5, 90°)	β Cephei (V) (2.5, 180°)	δ Scuti (V) (2.5, 90°)	Short-Period (V) (1.5, 90°)			
		0	30	0	30	0	30	0	40	0	15
$l=0$		1.9	3.0	0.15	0.16	0.33	0.34	0.30	0.30	145	45
1		1.2	1.6	0.09	0.09	0.36	0.37	0.33	0.33	170	55
2		0.9	1.1	0.08	0.07	0.38	0.40	0.36	0.36	225	65
3		3.2	3.4	0.33	0.32	0.51	0.50	0.49	0.49	400	95
4		0.33	0.31	0.03	0.03	0.14	0.18	0.34	0.34	95	40
5		1.6	1.7	0.31	0.30	0.36	0.33	0.33	0.33	85	15

$(\phi_{\lambda_2} - \phi_{\text{col}})$ in degrees										$(\phi_{\lambda_2} - \phi_{\text{RV}})$ in degrees									
$l=0$		0	0	0	0	0	0	0	0	0	90	90	90	90	90	90	90	90	90
1		0	0	0	0	0	0	0	0	0	90	90	90	90	90	90	90	90	90
2		0	0	0	0	0	0	0	0	0	90	90	90	90	90	90	90	90	90
3		0	0	0	0	0	0	0	0	0	90	90	90	90	90	90	90	90	90
4		180	180	180	180	180	180	180	180	180	90	90	90	90	90	90	90	90	90
5		0	0	0	0	0	0	0	0	0	90	90	90	90	90	90	90	90	90

for $l \leq 2$. Thus when $\psi_T = \pi$, as in the β Cephei test case, the geometric contribution to A_V changes the ratio (30) considerably as l shifts from 0 to 1 and then 2. For the δ Scuti and short-period Cepheid cases, on the other hand, because $\psi_T = \pi/2$, little change occurs in (30) as l changes from 0 through to 2.

For $l > 2$ the simplification of (12) which led to the approximate expression (30) breaks down. For $l \leq 2$ colours are dominated by the 'temperature' term in (12), $(\alpha_T(\lambda_1)b_{l\lambda_1} - \alpha_T(\lambda_2)b_{l\lambda_2})B$. However, by $l = 3$ the 'geometric' term $(2+l)(1-l)(b_{l\lambda_1} - b_{l\lambda_2})$ is substantial and by $l = 4$ it dominates in all test regimes. Thus by $l = 4$ it is the wavelength dependence of limb-darkening which is the chief source of colour variations in all test models. The geometric term in A_V is also the dominant one by $l = 4$.

Effects of the minimum uncertainties in $\alpha_T(\lambda)$ and $b_{l\lambda}$, discussed in Section 5.2, were examined. This examination confirmed that $A_{\text{col}}/A_{\lambda_2}$ values are not reliable l discriminants for any $l < 5$ in the β Cephei (B, V), δ Scuti or short-period Cepheid cases, quite apart from any observational errors. It is the $\alpha_T(\lambda)$ errors which are the problem for $l \leq 2$ and the $b_{l\lambda}$ errors which can make colours almost meaningless for $l \geq 3$. Only in the β Cephei (1500, V) case does the possibility of l discrimination remain in the face of these $\alpha_T(\lambda)$ and $b_{l\lambda}$ uncertainties. Even in this case discrimination is not possible for $l = 3$ and 5 because of overlap with lower l values and sensitivity of colours to $b_{l\lambda}$ errors. The effect on A_{1500-V}/A_V of varying (B, ψ_T) about the values in Table IV is shown in Figure 5 for $l \leq 2$. This affirms the mode discriminating ability of A_{1500-V}/A_V under wide (B, ψ_T) uncertainties, particularly if pressure effects are accounted for. The success of this A_{1500-V}/A_V as a mode discriminator has resulted from a combination of a large $(\alpha_T(\lambda_1) - \alpha_T(\lambda_2))$ value, which enables reliable colours to be obtained theoretically and observationally, together with a small $\alpha_T(\lambda_2)$ value, which enables the geometric term in (11) to affect A_V in a low l and $\psi_T = \pi$ situation. It is interesting to note that, had B values in the β Cephei test case been close to adiabatic ones, such discrimination would not have been possible.

The K/A_{λ_2} values in Table VI show an l dependence which is most marked in the β Cephei test case. As for $A_{\text{col}}/A_{\lambda_2}$, the reason for this is the effect of the geometric contribution to A_V when $\psi_T = \pi$. Errors in $\alpha_T(\lambda)$ and $b_{l\lambda}$ values alone do not destroy the mode discrimination possibilities of K/A_V . However, its usefulness is severely limited by the effects of uncertainties in R and B , and by the Section 4 problems related to the means by which radial velocity measurements are made.

The $(\phi_{\lambda_2} - \phi_{\text{col}})$ values for the δ Scuti and short-period Cepheid test cases in Table VI show a variation with l which, when minimum uncertainties in $\alpha_T(\lambda)$ and $b_{l\lambda}$ are allowed for, suggests that $l \leq 2$ and $l = 4$ modes may be identified by careful observation. The $l = 3$ and 5 modes are confused with smaller l . For $l \leq 2$, the effect of varying (B, ψ_T) about Table IV values is illustrated in Figure 6. This demonstrates that, provided pressure effects are allowed for, $(\phi_V - \phi_{B-V})$

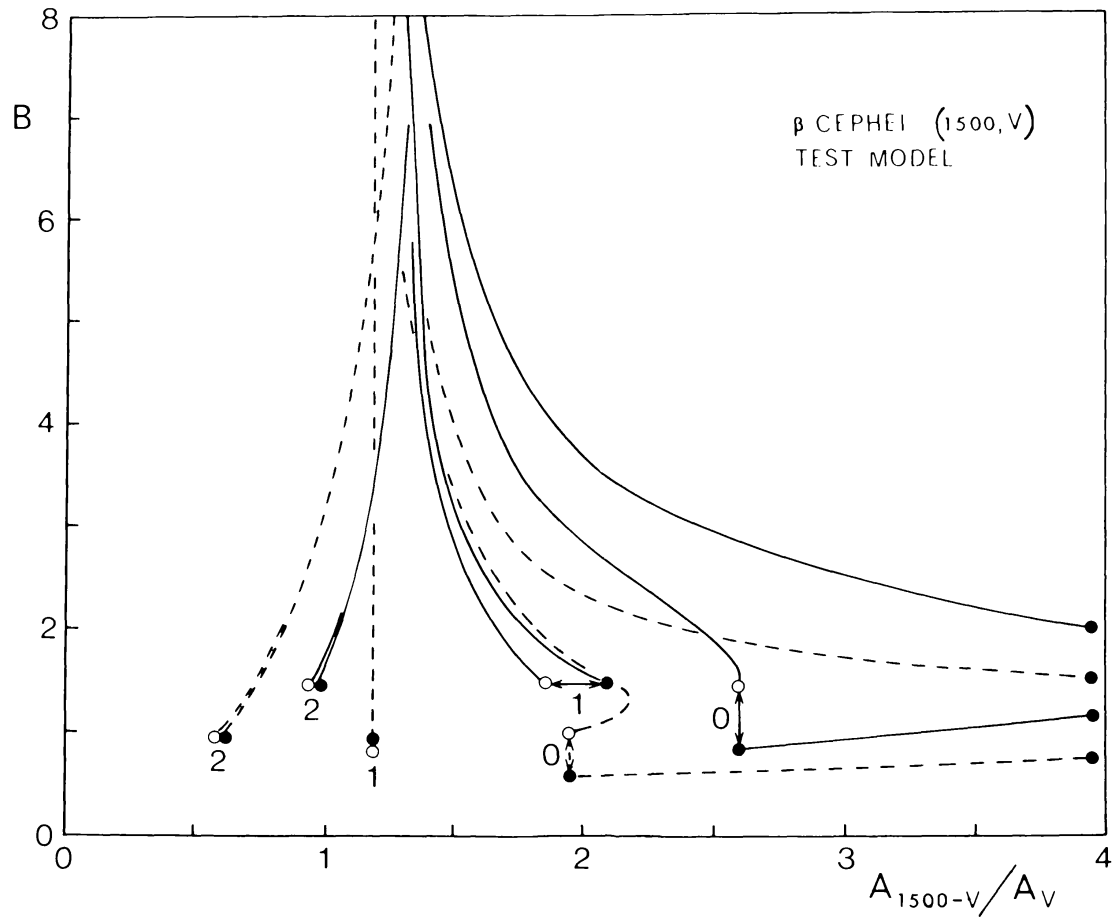


Fig. 5. A_{1500-V}/A_V versus (B, ψ_T) for the β Cephei test model with $l = 0, 1, 2$. Full lines are for $p^*C = 30$ and dashed lines are for $p^*C = 0$. Filled circles indicate $\psi_T = 180^\circ$ and open circles indicate $\psi_T = 150^\circ$ or 210° .

remains a useful discriminator in spite of substantial (B, ψ_T) uncertainties. However, $l \leq 2$ discrimination would not be possible if observed B were close to adiabatic values. In Table VII the values of $(\phi_V - \phi_{B-V})$ for the δ Scuti model are given for a range of $(\alpha_T(V)B, \psi_T)$ to enable direct comparison with Balona and Stobie (1980). Values for both $p^*C = 0$ and $p^*C = 40$ are tabulated. There was an indication from the Baade-Wesselink method of a pressure effect in the 1 Mon radial mode data. The $(\phi_V - \phi_{B-V})$ value for the principal mode in HD101065 also favours the existence of such an effect. Its identification by Kurtz (1980) with an $l = 2$ pulsation is achieved without the need for a very low B value if pressure effects are present.

For $l \leq 2$ the source of the discriminating ability in $(\phi_V - \phi_{B-V})$ is the geometric term in the light variations, as given by (11). In the absence of pressure effects, the phase of these light variations varies through ψ_T as l changes from 0 through to 2. The phase of the colour variations, on the other hand, is closely locked to ψ_T in the absence of pressure effects, because the

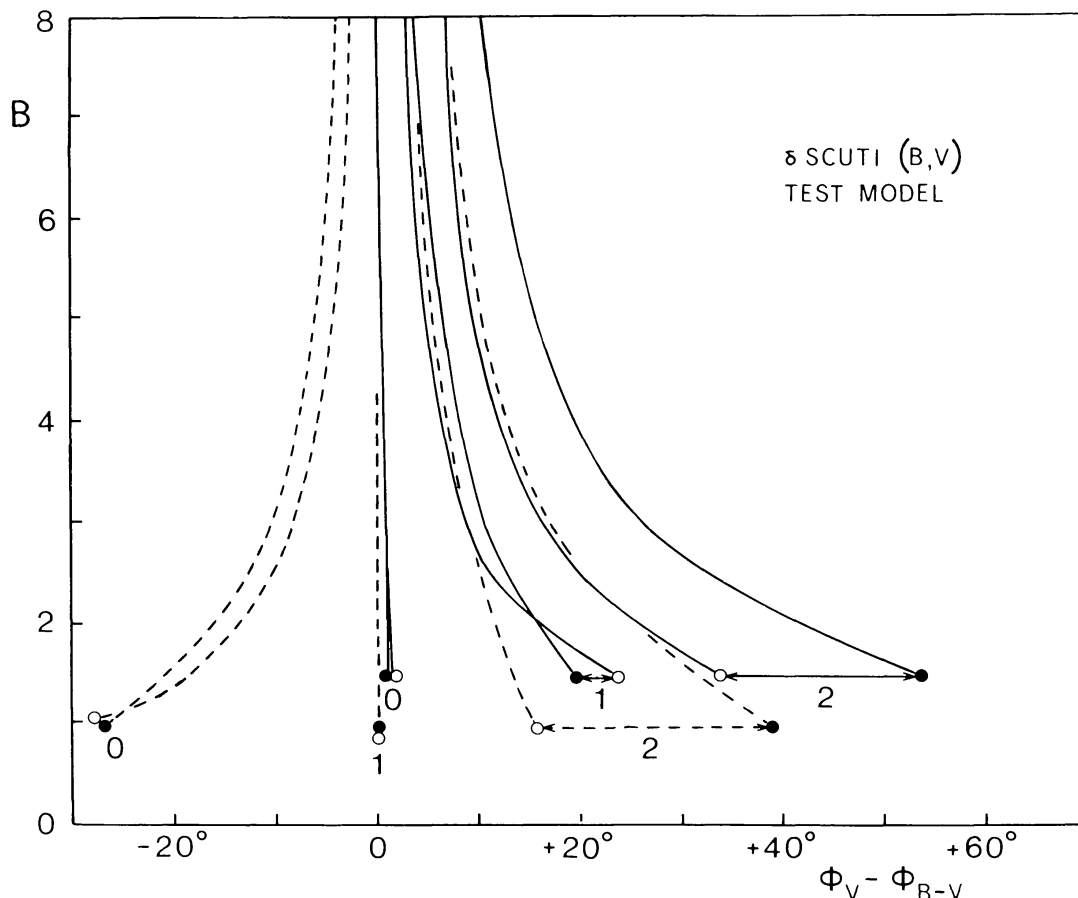


Fig. 6. $(\phi_V - \phi_{B-V})$ versus (B, ψ_T) for the δ Scuti test model with $l=0, 1, 2$. Full lines are for $p \cdot C = 40$ and dashed lines are for $p \cdot C = 0$. Filled circles indicate $\psi_T = 90^\circ$ and open circles indicate $\psi_T = 140^\circ$.

geometric term in (12) is either small or zero for $l \leq 2$. As in the case of a successful $A_{\text{col}}/A_{\lambda_2}$ discriminator, a desirable observational system here is one in which a large $(\alpha_T(\lambda_1) - \alpha_T(\lambda_2))$ for accurate theoretical and observational colours is combined with a small $\alpha_T(\lambda_2)$ value to provide large phase changes from (11) as l varies from 0 to 2. For $l \geq 3$ the geometric term in (12) and the associated limb-darkening uncertainties in $b_{l\lambda}$ begin to affect the $(\phi_{\lambda_2} - \phi_{\text{col}})$ values.

Values of $(\phi_{\lambda_2} - \phi_{RV})$ in Table VI at first sight suggest a mode discriminating ability similar to $(\phi_{\lambda_2} - \phi_{\text{col}})$, particularly in view of the fact that they are independent of errors in $b_{l\lambda}$ and insensitive to $\alpha_T(\lambda)$ errors. However, the discriminating ability is completely removed by uncertainties in ψ_T . The difference between the $(\phi_{\lambda_2} - \phi_{RV})$ case and that for $(\phi_{\lambda_2} - \phi_{\text{col}})$ is simply that, for $l \leq 2$, ϕ_{col} remains tied to ψ_T , whereas ϕ_{RV} does not.

The general conclusion drawn from all the preceding discussion is that $A_{\text{col}}/A_{\lambda_2}$ or $(\phi_{\lambda_2} - \phi_{\text{col}})$ are useful l discriminators depending on whether $\psi_T \sim \pi$ or $\sim \pi/2$. A similar conclusion is implicit in the work of Balona and Stobie (1979c). To this conclusion must be added the provisos that pressure effects be included and

TABLE VII
 $(\phi_V - \phi_{B-V})$, in degrees, for the δ Scuti test model

$\alpha_T(V)B$		5.0		10.0		15.0		20.0		25.0		30.0	
p^*C		0	40	0	40	0	40	0	40	0	40	0	40
$l = 0$	$\psi_T = 90$	-22	1	-11	1	-8	0	-5	0	-5	0	-4	0
	100	-23	1	-12	1	-8	0	-5	0	-5	0	-4	0
	120	-23	2	-11	1	-7	0	-5	0	-4	0	-3	0
	140	-20	2	-9	1	-6	0	-4	0	-3	0	-3	0
1	90	0	21	0	11	0	8	0	6	0	5	0	4
	100	0	23	0	12	0	8	0	6	0	5	0	4
	120	0	26	0	12	0	7	0	5	0	4	0	4
	140	0	26	0	10	0	6	0	4	0	3	0	3
2	90	32	55	19	30	13	20	10	15	8	12	7	10
	100	29	53	17	29	12	19	9	15	7	12	6	10
	120	21	47	13	25	10	17	8	13	6	10	5	9
	140	14	36	9	18	7	12	5	9	4	7	4	6
3	90	26	42	24	33	19	25	16	20	13	16	11	14
	100	22	36	21	29	17	22	14	18	12	15	10	13
	120	15	25	15	21	12	17	11	14	9	12	8	10
	140	10	16	9	14	8	11	7	9	6	8	5	7
4	90	147	150	118	124	96	103	80	86	68	75	58	64
	100	147	151	118	125	96	103	79	87	67	74	57	64
	120	150	154	120	129	95	107	76	88	63	74	53	63
	140	156	160	127	139	95	114	70	90	54	70	44	57
5	90	14	18	22	27	25	30	25	29	24	27	23	25
	100	13	16	19	23	21	25	21	25	20	23	19	22
	120	9	12	13	16	15	17	15	17	14	16	14	15
	140	6	8	9	10	9	11	9	11	9	10	9	10

that, while discrimination between modes with $l \leq 2$ is feasible, other arguments are required to rule out overlapping higher l modes. Use of these discriminators has the obvious advantage of not requiring radial velocity data, particularly in view of the difficulties discussed in Section 4 and those discussed by Smith (1980a) for multimode pulsators.

7. Conclusions

It has been demonstrated that, for nonradial pulsations with $l > 1$, the hypothesis of a single-valued relation between observed surface brightness and colour variations is not valid. Abandonment of this hypothesis we can determine a Baade-Wesselink formula which will work, in principle, for all l values. Examples from different variable star types suggest that practical use of this formula for $l \leq 2$ is possible, provided that pressure effects are included. It has also been demonstrated that use of radial velocity measurements to infer mean

observed surface velocity changes for nonradial modes requires specific evaluation of mean wavelengths of line profiles. Failure to do this may even affect $l = 2$ data and a multimode situation brings further complications. It is these radial velocity difficulties which are the principal problem in applying the Baade-Wesselink formula to higher l . Overlap of discriminants, particularly for $l = 3$, is a further problem.

Radial velocity difficulties are avoided by use of techniques relying on amplitudes and phases of light and colour data. These techniques should also successfully discriminate for $l \leq 2$. Improvements in model atmosphere limb-darkening, combined with availability of data at many wavelengths (e.g. Stamford and Watson, 1980), may extend this discrimination ability to higher l .

References

- Balona, L. A.: 1977, *Monthly Notices Roy. Astron. Soc.* **178**, 231.
 Balona, L. A. and Feast, M. W.: 1975, *Monthly Notices Roy. Astron. Soc.* **172**, 191.
 Balona, L. A. and Martin, W. L.: 1978, *Monthly Notices Roy. Astron. Soc.* **184**, 1.
 Balona, L. A. and Stobie, R. S.: 1979a, *Monthly Notices Roy. Astron. Soc.* **187**, 217.
 Balona, L. A. and Stobie, R. S.: 1979b, *Monthly Notices Roy. Astron. Soc.* **189**, 627.
 Balona, L. A. and Stobie, R. S.: 1979c, *Monthly Notices Roy. Astron. Soc.* **189**, 649.
 Balona, L. A. and Stobie, R. S.: 1979d, *Monthly Notices Roy. Astron. Soc.* **189**, 659.
 Balona, L. A. and Stobie, R. S.: 1980, *Monthly Notices Roy. Astron. Soc.* **190**, 931.
 Beeckmans, F. and Burger, M.: 1977, *Astron. Astrophys.* **61**, 815.
 Bell, R. A. and Gustafsson, B.: 1978, *Astron. Astrophys. Suppl.* **34**, 229.
 Breger, M.: 1975, *Astrophys. J.* **201**, 653.
 Breger, M.: 1979, *Publ. Astron. Soc. Pacific* **91**, 5.
 Breger, M. and Bregman, J. N.: 1975, *Astrophys. J.* **200**, 343.
 Breger, M., Hutchins, J., and Kuhl, L. V.: 1976, *Astrophys. J.* **210**, 163.
 Buta, J. R. and Smith, M. A.: 1979, *Astrophys. J.* **232**, 213.
 Campos, A. J. and Smith, M. A.: 1980, *Astrophys. J.* **238**, 250.
 Cox, A. N.: 1979, *Astrophys. J.* **229**, 212.
 Davey, W. R.: 1970, unpublished thesis, Univ. of Colorado, Boulder.
 Duval, P. and Karp, A. H.: 1978, *Astrophys. J.* **222**, 220.
 Dziembowski, W.: 1977, *Acta Astron.* **27**, 203.
 Jerzykiewicz, M.: 1978, *Acta Astron.* **28**, 465.
 Kurtz, D. W.: 1980, *Monthly Notices Roy. Astron. Soc.* **191**, 115.
 Kurucz, R. L.: 1969, in O. Gingerich (ed.) *Proc. Third Harvard Smithsonian Conf. on Stellar Atmospheres*.
 Kurucz, R. L.: 1979, *Astrophys. J. Suppl.* **40**, 1.
 Lucy, L. B.: 1976, *Astrophys. J.* **206**, 499.
 McGraw, J. T.: 1979, *Astrophys. J.* **229**, 203.
 McGraw, J. T.: 1978, *Proc. 4th Goddard/Los Alamos Conference, Current Problems in Stellar Pulsational Instabilities*.
 McNamara, D. H. and Feltz, K. A.: 1976, *Publ. Astron. Soc. Pacific* **88**, 510.
 Parsons, S. B.: 1972, *Astrophys. J.* **174**, 57.
 Robinson, E. L.: 1978, *Proc. 4th Goddard/Los Alamos Conference, Current Problems in Stellar Pulsational Instabilities*.
 Saio, H. and Cox, J. P.: 1980, *Astrophys. J.* **236**, 549.
 Shobbrook, R. R.: 1978, *Proc. IAU Coll.* **46**, Univ. of Waikato, New Zealand, p. 512.
 Smith, M. A.: 1978, *Astrophys. J.* **224**, 927.
 Smith, M. A.: 1980a, *Astrophys. J.* **240**, 149.

- Smith, M. A.: 1980b, *Astrophys. J.* (in press).
- Stamford, P. A. and Watson, R. D.: 1977, *Monthly Notices Roy. Astron. Soc.* **180**, 551.
- Stamford, P. A. and Watson, R. D.: 1978, *Proc. IAU Coll.* **46**, Univ. of Waikato, New Zealand, p. 504.
- Stamford, P. A. and Watson, R. D.: 1980, *Acta Astron.* **30**, 193.
- Stellingwerf, R. F.: 1979, *Astrophys. J.* **227**, 935.
- Watson, R. D.: 1971, *Astrophys. J.* **170**, 345.